

Lecture 1

1. Probability:

definition of probability: $P(A)$: probability of event A .

$$P(A) = \frac{N_A}{N}$$

N_A : number of occurrences of A

N : number of all possible outcomes

2. conditional Probability:

$P(A|B)$: The probability of event A given that event B is true.

$$P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(A, B)}{P(B)} \quad P(B) \neq 0$$

$$\Rightarrow P(A, B) = P(A|B) P(B) = P(B|A) P(A)$$

3. Total Probability Theorem:

If $\{A_1, \dots, A_n\}$ is a partition of A , B is arbitrary event:

$$P(A) = P(A_1, A_2, \dots, A_n) = P(A_1) + \dots + P(A_n), \quad P(A_i, A_j) = 0$$

$$P(B) = P(B|A_1) P(A_1) + P(B|A_2) P(A_2) + \dots + P(B|A_n) P(A_n)$$

$$= \sum_i P(B|A_i) P(A_i)$$

$$P(B) = P(B, A_1) + P(B, A_2) + \dots + P(B, A_n)$$

$$= P(B|A_1) P(A_1) + \dots + P(B|A_n) P(A_n)$$

$$= \sum P(B|A_i) P(A_i)$$

4. Bayes' Theorem:

$$P(A_i|B) = \frac{P(B|A_i) P(A_i)}{P(B)}$$

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} = \frac{P(B|A_i) P(A_i)}{\sum_i P(B|A_i) P(A_i)}$$

Independence:

$$5. P(A \text{ and } B) = P(A) + P(B) - P(A \text{ or } B)$$

$$P(A|B) = P(A|B) P(B) = P(A) P(B) \\ = P(B|A) P(A) = P(B) P(A)$$

Random Variables:

A random variable defines a function that quantifies an experiment.

For example: $S = \{s_1, \dots, s_n\}$, $x: S \rightarrow \mathbb{R}$, $x(s_i) = x_i$

$$s_i \Rightarrow t_i, \quad x_i = v_i \quad V(t_i) = v_i$$

$$P(t_i) = p_i$$

Definition: Distribution: The distribution of a random variable:

$$F_X(x) = P(X \leq x)$$

$P(X \leq x)$ represents the probability that x has values lower than

Probability density Function (pdf):

$$p(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x) - F_X(x-\epsilon)}{\epsilon} = \frac{dF_X(x)}{dx} = \frac{dP(X \leq x)}{dx}$$

$$\text{Hence: } P(X \leq x) = \int_{-\infty}^x p(\epsilon) d\epsilon$$

$$P(X \leq \infty) = 1 = \int_{-\infty}^{+\infty} p(\epsilon) d\epsilon$$

is normal.

$$P(x_1 \leq x < x_2) = \int_{x_1}^{x_2} p(x) dx$$

example distributions:

Gaussian distributions: $x \sim N(\mu, \sigma) \Leftrightarrow p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Uniform distributions:

$$p(x) = \begin{cases} 0 & x < x_1 \\ \frac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$$

conditional distribution:

$$P(x \leq x | A) = \frac{P(x \leq x, A)}{P(A)}$$

conditional probability density function:

$$p(x|A) = \frac{dP(x \leq x | A)}{dx}$$

The Total probability theorem for pdf:

For: $A = \{A_1, \dots, A_n\}$, $A_1, \dots, A_n$ are exclusive.

$$P(x \leq x | A) = P(x \leq x | A_1)P(A_1) + \dots + P(x \leq x | A_n)P(A_n)$$

$$p(x) = p(x|A_1)P(A_1) + \dots + p(x|A_n)P(A_n)$$

Example:

$$p(x) = \frac{dP(x \leq x)}{dx} \approx \frac{P(x \leq x + \Delta x) - P(x \leq x)}{\Delta x}$$

$$P(x=x) = P(x \leq x + \Delta x) - P(x \leq x) = p(x) \cdot \Delta x$$

$$P(x=x|A) = p(x|A) \cdot \Delta x$$

$$P(A|x=x) = \frac{P(A, x=x)}{P(x=x)} = \frac{P(x=x|A)P(A)}{P(x=x)} = \frac{p(x|A)}{p(x)} \cdot P(A)$$

$$\Rightarrow P(A|x=x)P(x) = P(x|A)P(A)$$

continuous theorem:

$$P(A) = \int_{-\infty}^{+\infty} P(A|x=x)P(x)dx$$

continuous Bayes' theorem:

$$P(x|A) = \frac{P(A|x=x) \cdot P(x)}{P(A)} = \frac{P(A|x=x)P(x)}{\int_{-\infty}^{+\infty} P(A|x=x)P(x)dx}$$

Mean: The expected value or mean of a continuous variable is:

$$m_x = E(x) = \int_{-\infty}^{+\infty} xP(x)dx$$

Conditional Mean:

$$m_{x|A} = E(x|A) = \int_{-\infty}^{+\infty} xP(x|A)dx$$

Variance: $\sigma_x^2 = E((x-m_x)^2) = \int_{-\infty}^{+\infty} (x-m_x)^2 P(x)dx$

$$= E(x^2) - (E(x))^2$$

conditional variance:

$$\sigma_{x|A}^2 = E\{(x-m_{x|A})^2|A\} = \int_{-\infty}^{+\infty} (x-m_{x|A})^2 P(x|A)dx$$

A set of random variables:

Joint probability density function:

$$x_1, \dots, x_n, \quad p(x_1, \dots, x_n) = \frac{dF(x_1, \dots, x_n)}{dx_1 \dots dx_n}$$

$$dx_1, \dots, dx_n$$

$$\int_{-\infty}^{+\infty} \dots \int_{-\infty}^{+\infty} p(x_1, \dots, x_n) dx_1, \dots, dx_n = 1$$

if x_i, x_j , are independent, $p(x_i, x_j) = p(x_i)p(x_j)$

$$\text{for: } y = \alpha x_i + \beta x_j \Rightarrow E(y) = \alpha E(x_i) + \beta E(x_j)$$

$$\begin{aligned} \text{covariance for } \sigma_{ij}: \quad \sigma_{ij}^2 &= E((x_i - m_i)(x_j - m_j)) \\ &= E(x_i x_j) - E(x_i) E(x_j) \end{aligned}$$

$$\text{Hence: } \sigma_{ij} = 0, \quad E(x_i x_j) = E(x_i) E(x_j)$$

they are uncorrelated.

$$\textcircled{1} \text{ If } x_i, x_j \text{ are independent, } p(x_i, x_j) = p(x_i)p(x_j) \Rightarrow E(x_i x_j) = E(x_i) E(x_j)$$

$\textcircled{2}$ If x_i, x_j are uncorrelated, are they independent?

$\textcircled{3}$ if x_i, x_j subject to Gaussian distribution, independent $\stackrel{?}{=}$ uncorrelated?

Then random vector

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

All the element are random variables

$$E(X) \triangleq m_X$$

$$P = E((X - m_X)(X - m_X)^T) = E(X X^T) - E(X) E(X^T)$$

$$P = \begin{bmatrix} P_{11} & \dots & P_{1n} \\ \vdots & & \vdots \\ P_{n1} & \dots & P_{nn} \end{bmatrix}$$

$$P_{ii} = E(x_i - m_{x_i})^2 = \int_{-\infty}^{+\infty} (x_i - m_{x_i})^2 p(x_i) dx_i$$

$$P_{ij} = E((x_i - m_{x_i})(x_j - m_{x_j}))$$

$$\begin{cases} P \text{ is symmetric Matrix} \\ P \text{ is positive definite Matrix.} \end{cases} = P_{ji}$$

Gaussian vector pdf: $p(x) = \frac{1}{(2\pi)^{\frac{n}{2}} \sqrt{|P|}} e^{-\frac{1}{2}(x-\mu_x)^T P^{-1} (x-\mu_x)}$

$$\begin{cases} |P| \text{ is the determinat of } P \\ \mu_x = E(x) \\ P = E((x-\mu_x)(x-\mu_x)^T) \end{cases}$$

A homework: $x = [x_1, \dots, x_n]^T$

$$p(x) = N(\mu_x, P_{xx})$$

define $q = (x-\mu_x)^T P_{xx}^{-1} (x-\mu_x)$ $y = P_{xx}^{-\frac{1}{2}} (x-\mu_x)$

$$= y^T y = \sum y_i^2$$

what's mean and covariance for q ?

maximum likely hood and max. posterior estimator:

$$\begin{cases} \max p(z|x) : \text{MLE} & p(z|x) \\ \max p(x|z) : \text{MPE} & p(x|z) = \frac{p(z|x)p(x)}{p(z)} \end{cases}$$

Gaussian distribution:

$$z_i = h_i(x) + n_i \Rightarrow \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} h_1(x) \\ h_2(x) \\ \vdots \\ h_n(x) \end{bmatrix} + \begin{bmatrix} n_1 \\ \vdots \\ n_n \end{bmatrix}$$

$$z = h(x) + n$$

$$\max_x p(z|x) = \max_x (p(z_1, \dots, z_n|x))$$

$$= \max_x \left(\frac{1}{(2\pi)^n |P|} \exp(-(z-h(x))^T P^{-1} (z-h(x))) \right)$$

$$\max_x p(z|x) \Rightarrow \min_x (z-h(x))^T P^{-1} (z-h(x)) = C$$

$$z = h(x) + n = h(\hat{x}) + H(x-\hat{x}) + n$$

$$z - h(\hat{x}) = H(x-\hat{x}) + n \quad \tilde{z} = z - h(\hat{x}) \quad x - \hat{x} = \tilde{x}$$

$$\min C \approx \min (\tilde{z} - H\tilde{x})^T P^{-1} (\tilde{z} - H\tilde{x})$$

$$= \min (\tilde{z}^T \tilde{z} - 2 \tilde{z}^T P^{-1} H \tilde{x} + \tilde{x}^T H^T P^{-1} H \tilde{x})$$

$$H^T P^{-1} H \tilde{x} = H^T P^{-1} \tilde{z} \Rightarrow x^{\oplus} = \hat{x} + \tilde{x} = \hat{x} + (H^T P^{-1} H)^{-1} H^T P^{-1} \tilde{z}$$

This is the classical weighted least squares

What's the covariance of the new estimator: $x^{\oplus}$?

What's the MP?

$$\begin{cases} X_p = x + n_p \\ z_i = h_i(x) + n_i \quad i=1, \dots, n \end{cases}$$

Then, what's the solution for x of $\max p(x|z)$

optimization: $\begin{cases} \text{Gaussian Newton (GN)} \\ \text{Levenberg marquardt (LM)} \end{cases}$

... ..

Linear system:

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases} \quad \begin{cases} x: \text{is the state} \\ u: \text{input} \\ y: \text{output} \end{cases}$$

This is a continuous-time system:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ y_{k+1} = Cx_{k+1} + Du_{k+1} \end{cases} \quad \text{This is a discrete-time system.}$$

If A, B, C, D are constant, linear time invariant system (LTI)

If A, B, C, D are depend on time, linear time variant system (LTV)

Let's start from a simple system:

$$\begin{cases} \dot{x} = Ax, \text{ with } x(t_0) = x_0 \\ y = Cx \end{cases}$$

Solution for $\dot{x} = Ax$ with $x(t_0) = x_0$

$$\dot{x} - Ax = 0 \quad \frac{d(e^{-At}x)}{dt} = e^{-At}\dot{x} + e^{-At}(-A)x = e^{-At}(\dot{x} - Ax) = 0$$

$$\int_0^t \frac{d(e^{-A\tau}x)}{d\tau} = 0 = e^{-At}x \Big|_0^t = e^{-At}x(t) - e^{-A \cdot 0}x(t_0) = e^{-At}x(t) - x(t_0)$$

$$x(t) = e^{At}x_0 \quad x(t) = \Phi(t, 0)x_0 \quad \Phi(t, 0) = e^{At} \text{ state transition matrix.}$$

$$y = C \cdot e^{At}x_0$$

$$\text{Hence: the final solution: } \begin{cases} x(t) = \Phi(t, 0)x_0 \\ y = C\Phi(t, 0)x_0 \end{cases}, \quad \Phi(t, 0) = e^{At}$$

What's the property of state transition matrix?

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0)$$

$$\frac{d}{dt} \Phi(t, 0) = A \Phi(t, 0)$$

$$\dot{x} = Ax$$

$$x = \Phi(t, 0) x_0$$

$$\dot{x} = \frac{d}{dt} \Phi(t, 0) x_0 = A x = A \Phi(t, 0) x_0$$

Find the correct prove for this.

Then: with $x(t) = \Phi(t, 0) x_0$

$$y = Cx$$

$$t_0: \quad x_0 \quad y_0 = Cx_0$$

$$y_0 = Cx_0$$

$$t_1: \quad x_1 = \Phi(1, 0) x_0 \quad y_1 = Cx_1$$

$$y_1 = Cx_1 = C \Phi(1, 0) x_0$$

$$t_2: \quad x_2 = \Phi(2, 1) x_1 \quad y_2 = Cx_2$$

$$y_2 = Cx_2 = C \Phi(2, 1) x_1$$

$$\vdots$$

$$\vdots$$

$$t_n: \quad x_n = \Phi(n, 0) x_0 \quad y_n = Cx_n$$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} C \Phi(1, 0) \\ \vdots \\ C \Phi(n, 0) \end{bmatrix} x_0$$

①

Now let's consider the whole system:

$$\dot{x} = A(t)x + B(t)u(t)$$

solve this linear system.

$$y = C(t)x + D(t)u(t)$$

$$\dot{x} = A(t)x + B(t)u(t) \Rightarrow \dot{x} - A(t)x = B(t)u(t)$$

$$\begin{aligned} \frac{d \left[e^{-\int_0^t A(\tau) d\tau} x \right]}{dt} &= e^{-\int_0^t A(\tau) d\tau} \dot{x} - e^{-\int_0^t A(\tau) d\tau} A(t) \cdot x \\ &= e^{-\int_0^t A(\tau) d\tau} (\dot{x} - A(t)x) \\ &= e^{-\int_0^t A(\tau) d\tau} B(t)u(t) \end{aligned}$$

$$e^{-\int_0^t A(\tau) d\tau} x \Big|_0^t = e^{-\int_0^t A(\tau) d\tau} x(t) - e^{-\int_0^0 A(\tau) d\tau} x_0$$

$$= e^{-\int_0^t A(\tau) d\tau} x(t) - x_0$$

$$= \int_0^t e^{-\int_0^{\tau} A(s) ds} B(\tau) u(\tau) d\tau$$

$$x(t) = \underbrace{e^{\int_0^t A(\tau) d\tau}}_{\Phi(t,0)} x_0 + \int_0^t \underbrace{e^{\int_0^{t-\tau} A(s) ds}}_{\Phi(t,\tau)} B(\tau) u(\tau) d\tau$$

$$x(t) = \Phi(t,0) x_0 + \int_0^t \Phi(t,\tau) B(\tau) u(\tau) d\tau$$

$$y = C(t) x(t) + D(t) u(t)$$

$$= C(t) \left(\Phi(t,0) x_0 + \int_0^t \Phi(t,\tau) B(\tau) u(\tau) d\tau \right) + D(t) u(t)$$

This is one of the most important linear system properties.

Home work 1:

$$\sigma^2 = E\{(x - m_x)^2\} = E\{(x - m_x)^2\}$$

$$= \int_{-\infty}^{+\infty} (x - m_x)^2 p(x) dx = E(x^2) - E^2(x)$$

$$E(x^2) = \int_{-\infty}^{+\infty} x^2 p(x) dx$$

$$E(x) = \int_{-\infty}^{+\infty} x p(x) dx$$

$$E^2(x) = \int_{-\infty}^{+\infty} x_1 p(x_1) dx_1 \int_{-\infty}^{+\infty} x_2 p(x_2) dx_2$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_1 x_2 p(x_1) p(x_2) dx_1 dx_2$$

$$\text{if } E(x^2) = E^2(x) \Rightarrow \begin{cases} p(x) = 0 \\ \text{or:} \\ x \text{ always} = m_x \end{cases} \Rightarrow \text{This a very special case}$$

Home work 2:

If x is random vector, then

$$\sigma^2 = E\{(x - m_x)(x - m_x)^T\}$$

$$V_{xx} = E\{xx^T - 2x\mu_x^T + \mu_x\mu_x^T\}$$

$$= E\{xx^T - 2x\mu_x^T + \mu_x\mu_x^T\}$$

what's the details inside?

$$= \int_{-\infty}^{+\infty} xx^T p(x) dx - 2 \int_{-\infty}^{+\infty} x p(x) dx \mu_x^T + \mu_x \mu_x^T$$

$$= E\{xx^T\} - 2\mu_x \mu_x^T + \mu_x \mu_x^T$$

$$= E\{xx^T\} - E\{x\} E\{x\}^T$$

Homework 3:

If x_i, x_j are two random variables

$$\sigma_{ij} = E\{(x_i - \mu_i)(x_j - \mu_j)\}$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x_i - \mu_i)(x_j - \mu_j) p(x_i, x_j) dx_i dx_j$$

$$= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_i x_j p(x_i, x_j) dx_i dx_j - \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_i \mu_j p(x_i, x_j) dx_i dx_j$$

$$- \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_j \mu_i p(x_i, x_j) dx_i dx_j + \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \mu_i \mu_j p(x_i, x_j) dx_i dx_j$$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} p(x_i, x_j) dx_i dx_j = 1$$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_j \mu_i p(x_i, x_j) dx_i dx_j = \mu_i \int_{-\infty}^{+\infty} x_j \left[\int_{-\infty}^{+\infty} p(x_i, x_j) dx_i \right] dx_j$$

$$= \mu_i \int_{-\infty}^{+\infty} x_j p(x_j) dx_j$$

$$= \mu_i \mu_j$$

$$= E\{x_i x_j\} - 2\mu_i \mu_j + \mu_i \mu_j$$

$$= E\{x_i x_j\} - E\{x_i\} \cdot E\{x_j\}$$

Homework 4:

if independent $\Rightarrow$ uncorrelated

① if independent

$$p(x_i, x_j) = p(x_i) p(x_j)$$

$$\begin{aligned} \Rightarrow E\{x_i x_j\} &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_i x_j p(x_i, x_j) dx_i dx_j \\ &= \int_{-\infty}^{+\infty} x_i p(x_i) dx_i \cdot \int_{-\infty}^{+\infty} x_j p(x_j) dx_j \\ &= E(x_i) E(x_j) \end{aligned}$$

② if uncorrelated $\stackrel{?}{\Rightarrow}$ independent:

$$E\{x_i x_j\} = E\{x_i\} E\{x_j\} = 0$$

$$\Rightarrow \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_i x_j p(x_i, x_j) dx_i dx_j - \int_{-\infty}^{+\infty} x_i p(x_i) dx_i \int_{-\infty}^{+\infty} x_j p(x_j) dx_j = 0$$

$$\Rightarrow \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x_i x_j (p(x_i, x_j) - p(x_i) p(x_j)) dx_i dx_j = 0 \Rightarrow p(x_i, x_j) = p(x_i) p(x_j)$$

③ Gaussian variables

$$x_i \sim N(m_i, \sigma_i^2) \quad p(x_i) = \frac{1}{\sigma_i} e^{-\frac{(x_i - m_i)^2}{2\sigma_i^2}}$$

$$x_j \sim N(m_j, \sigma_j^2) \quad p(x_j) = \frac{1}{\sigma_j} e^{-\frac{(x_j - m_j)^2}{2\sigma_j^2}}$$

$$\begin{aligned} p(x_i) p(x_j) &= \frac{1}{\sigma_i} \cdot \frac{1}{\sigma_j} e^{-\left\{ \frac{(x_i - m_i)^2}{2\sigma_i^2} + \frac{(x_j - m_j)^2}{2\sigma_j^2} \right\}} \\ &= \frac{1}{\sigma_i \sigma_j} e^{-\left\{ \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^T \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix} \right\}} \end{aligned}$$

$$\begin{bmatrix} x_i \\ x_j \end{bmatrix} \sim N\left(\begin{bmatrix} m_i \\ m_j \end{bmatrix}, P_{xx}\right) \quad P_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$$

$$\begin{aligned} p(x_i, x_j) &= \frac{1}{r} e^{-\left\{ \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^T P_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix} \right\}} \\ &= \frac{1}{r} e^{-\left\{ \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^T \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix} \right\}} \end{aligned}$$

if $\sigma_j = \sigma_{\hat{j}} = 0$, then:

$$p(x_i, x_j) = p(x_i) p(x_j)$$

Here: independent $\Leftrightarrow$ uncorrelated for Gaussian variables

Homework 5:

ch2 distribution

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad p(x) = N(\mu_x, \Sigma_x)$$

$$q = (x - m_x)^T P_{xx}^{-1} (x - m_x)$$

$$P_{XX} = LL^T = U^T U$$

$$P_{XX}^T = L^{-T} L^T = U^T U^{-T}$$

$$= (x - m_x)^T L^T L^{-1} (x - m_x)$$

$$= y^T y$$

For $y = L^{-1}(x - m_x)$

$$m_y = E\{y\} = \int_{-\infty}^{+\infty} y p(x) dx = \int_{-\infty}^{+\infty} L^{-1}(x - m_x) p(x) dx$$

$$= L^{-1} \left(\int_{-\infty}^{+\infty} x f(x) dx - M_x \right) = 0$$

$$P_{yy} = E \{ (y - m_y)(y - m_y)^T \} = E \{ yy^T \}$$

$$= E\{L^T(x - \mu_x)(x - \mu_x)^T L^T\} = L^T E\{(x - \mu_x)(x - \mu_x)^T\} L^T$$

$$= L^{-1} P_{xx} L^{-T} = L^{-1} L L^T L^{-T} = I_n = \begin{bmatrix} 1 & & \\ & 0 & \\ & & 1 \end{bmatrix}^{P_{xx}}$$

Hence:

$$E(q) = E(y^T y) = E\left\{\sum_i y_i^2\right\} = n$$

$$x^2 - 1 \quad \text{or} \quad x^2 - 1 = x^2 - 1$$

$$099 = E\{[q - m_q]^2\} = E\{(\sum y_i - n)^2\}$$

$$= E\left\{\left(\sum_i (y_i^2 - 1)\right)^2\right\}$$

$$= E\left\{\sum_i (y_i^2 - 1)^2\right\} \quad (\text{due to } E\{y_i^2\} = 1)$$

$$= \sum_i E\{(y_i^2 - 1)^2\}$$

$$= \sum_i E\{y_i^4 - 2y_i^2 + 1\}$$

$$= \sum_i \left\{ \underbrace{E(y_i^4)}_3 - 2 \underbrace{E(y_i^2)}_{=1} + 1 \right\} = \sum_i 2 = 2n$$

$$E(y_i^4) = \int_{-\infty}^{+\infty} y_i^4 p(y_i) dy_i \quad p(y_i) = \frac{1}{\sqrt{2}} e^{-\frac{y_i^2}{2}}$$

$$\frac{d}{dy_i}(uv) = u \cdot \frac{dv}{dy_i} + v \frac{du}{dy_i}$$

$$\int_{-\infty}^{+\infty} \frac{d}{dy_i}(uv) = \int_{-\infty}^{+\infty} u \frac{dv}{dy_i} + \int_{-\infty}^{+\infty} v \frac{du}{dy_i}$$

$$uv \Big|_{-\infty}^{+\infty} = \int_{-\infty}^{+\infty} u \frac{dv}{dy_i} + \int_{-\infty}^{+\infty} v \frac{du}{dy_i}$$

$$\int_{-\infty}^{+\infty} u \frac{dv}{dy_i} = uv \Big|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty} v \frac{du}{dy_i}$$

$$\int_{-\infty}^{+\infty} y_i^4 p(y_i) dy_i = \int_{-\infty}^{+\infty} \underbrace{y_i^3}_u \cdot \underbrace{y_i e^{-\frac{y_i^2}{2}}}_{dv} dy_i$$

$$u = y_i^3$$

$$v = -e^{-\frac{y_i^2}{2}}$$

$$= -y_i^3 e^{-\frac{y_i^2}{2}} \Big|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} e^{-\frac{y_i^2}{2}} dy_i^3$$

$$= 0 + \int_{-\infty}^{+\infty} 3y_i^2 e^{-\frac{y_i^2}{2}} dy_i$$

$$= 3 E(y_i^2) = 5$$